



CITYFARM

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AND JACQUE DE BOEHLER**

STAKEHOLDERS AND INITIAL BRIEF

Stakeholders

- Windmill Hill City Farm
- Hartcliffe City Farm
- St Werburghs City Farm

Brief

- Log livestock data (birth, vaccines, enclosure)
- Keep track of tasks and events for each animal/enclosure
- Simplify existing system of messy excel spreadsheet and outlook calendar
- Move animals from one farm to another

USER STORIES

- As a livestock manager, I want to view animal capacities at different farms, so that I can quickly check if there's space before moving an animal. (This avoids overcrowding and ensures animal welfare.)
- As a livestock manager, I want to create recurring vaccination events for new vaccines, so that I can keep track of updated schedules. (This ensures animal health and avoids missed vaccinations.)
- As a livestock manager, I want to see a combined calendar for all farms, so that I can avoid switching between them. (This saves time and simplifies animal management.)
- As an animal breeder, I want to securely trade animals with other farms within the app, so that I can facilitate collaboration and resource sharing.
- As a livestock manager, I want to access animal data and inventory on my phone during power outages, so that I can ensure animal well-being even when computers are unavailable.

DEMO



cityfarm.murraygrove.es

TECH STACK

- React as a frontend, interfaces with the backend client-side
- Spring Boot on the backend, providing a REST API
- MongoDB as the database
- Microsoft Auth (simply checks the domain name, no individual users)
- Simple Docker compose file to bring it up

ISSUES WE FACED IN TB1

- Not using the Kanban board to sync work up efficiently meaning people were doing the same work simultaneously
- Confusion on our tech stack choices
- Realising we designed the backend in a way where it can't support a feature we need

CLIENT FEEDBACK - TB1

- Different colors for identifying different farms for events
- Shared calendar between all farms
- Attaching animals/enclosures to events
- Different tabs for different farms on each page
- Events with checkboxes to signify completeness

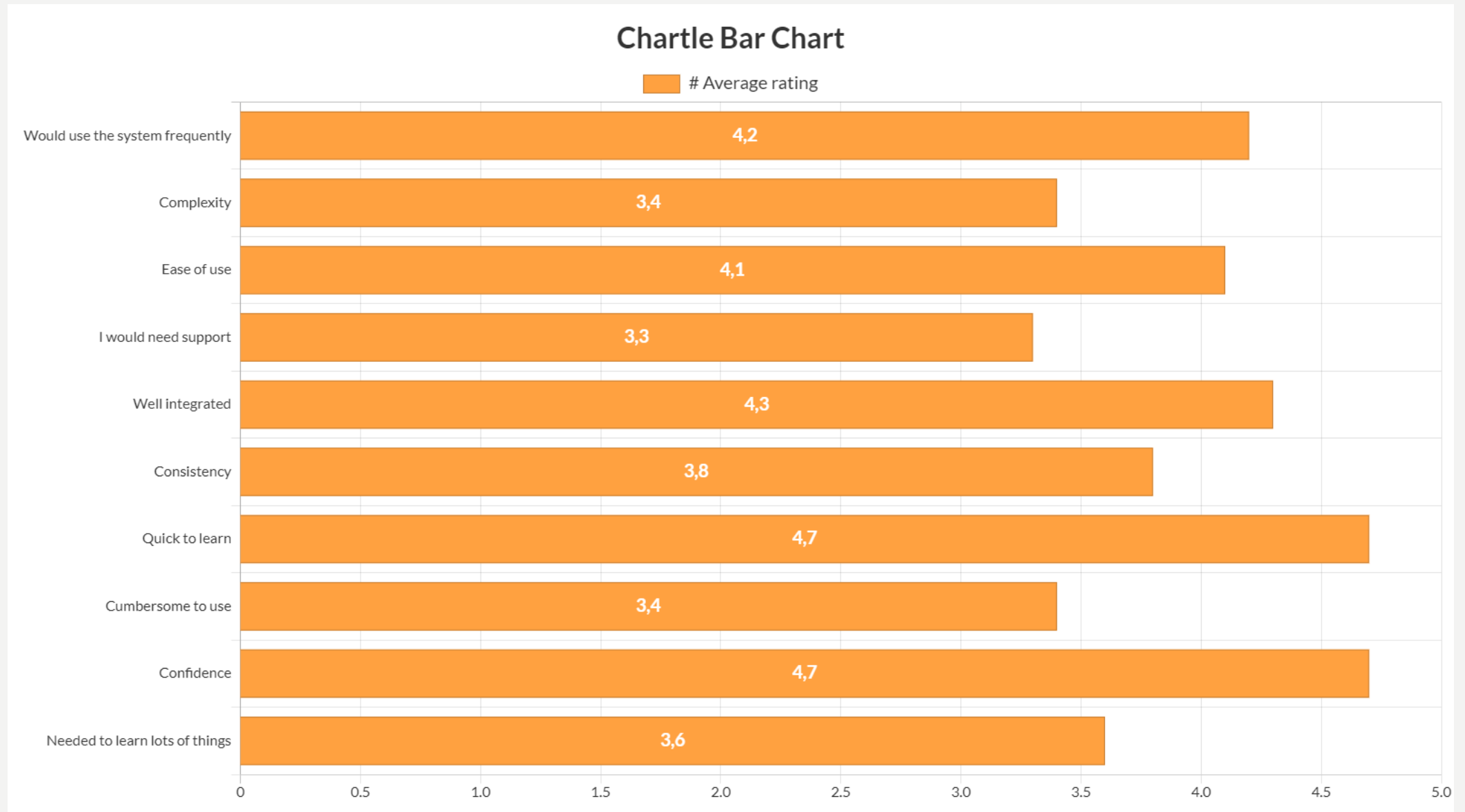
TB2 PLAN - COMPLETED

- Re-work the backend to allow users to create new animal types and attributes
- Fully integrate frontend with backend capabilities
- Frontend quality of life changes such as fields referencing other animals being links to those animals
- Login system to integrate with their existing Microsoft accounts
- Support for multiple farms

CLIENT FEEDBACK - TB2

- Sex isn't male/female, it's male/female/castrated - meant changing from a bool on the backend

TESTING DAY RESULTS - NORMALISED



TESTING DAY – WRITTEN FEEDBACK

For a farmer or livestock owner it seems quite useful, bit difficult for me as I needed to be told the different fields, but apart from that it was quite good

*It's not intuitively clear that hovering over an animal name gives more information about it, giving it a different style (like a hyperlink compared to standard text) would make it clearer
Schema seems like a weird name for a farmer using a template, maybe that should be a word more clearly related to the subject matter.*

you should be able to see children as well as parents!!!!!!

Our main takeaway from the first two pieces of feedback was that there was an accessibility issue, which we also saw was evident with the day and telling people how to use it, this led to us creating a home and help page as well as an icon to help with each individual page.

TESTING DAY RESPONSE

Home
Calendar
Livestock
Enclosures
Animal Types

Animal Types

Create New Animal Type

Data Type

This displays a page to create custom animal types, in case any new animals are to be added, once the type name is filled out, any number of properties can be added that pertain to that animal. Once an animal type is added, the creation of animals in the livestock table will be updated to include that.

Existing Animal Types

Help

On the left is the side menu, this contains various ways to manage livestock across the three farms. Below are instructions and help to navigate your way around the application, click the headers to reveal their content.

Calendar

Animal Table

This table displays all the animals, each one has 5 main attributes, father, mother, name, type and sex. This can be searched. This table is clickable and will take you to a single animal view page that will display more information about the animal, including move an animal's fields, select it on the grid and then press the move farm buttons below the table. The add animal button is defined in the animal types page and all the fields can be assigned.

Enclosures

Every enclosure has animals assigned to it. Click on an enclosure to expand it, you can edit them by clicking the edit button.

Animal Types

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Murray

Sex: Male

Species: Goat

Father: Unregistered

Mother: Unregistered

Farm: St Werburghs

Notes: Murray is a real one.

Children:

Ricky

Bobby

Home

Calendar

Livestock

Enclosures

Animal Types



PROCESS

- Very flexible development process
- Each of us pick and choose from issues on the Kanban
- Often merging lots of work around the same time
- Meet every week to consolidate work
- Communicate over WhatsApp as well as on GitHub

ISSUES FACED IN TB2

- Re-working of livestock system to support custom animals took a while
- Ran out of Azure credits
- Some communication issues meant some PRs took longer than they should've to complete and merge
- Kanban board problems with there being an overflow of open issues that no-one was picking up, many of which weren't clear at all as to what needed to be done towards them



QUESTIONS?